MIL-H-5606D
26 January 1978
SUPERSEDING
MIL-H-5606C
30 September 1971

MILITARY SPECIFICATION

HYDRAULIC FLUID, PETROLEUM BASE; AIRCRAFT, MISSILE, AND ORDNANCE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a petroleum base hydraulic fluid for use in the -54 to +71°C (-65 to +160°F) temperature range in open systems and up to +135°C (+275°F) in closed, airless systems. This fluid is identified by Military Symbol OHA and NATO Code Number H-515 (see 6.5).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

FEDERAL

TT-T-656	-Tricresyl Phosphate
PPP-C-96	-Cans, Metal, 28 Gage and Lighter
PPP-P-420	-Plugs and Flanges (For Drum Closures)
PPP-P-704	-Pails, Metal: (Shipping, Steel, 1 through 12 Gallon)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9150

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MILITARY

MIL-H-6083	-Hydraulic Fluid, Petroleum Base, for Preservation and Testing
MIL-H-46170	-Hydraulic Fluid, Rust inhibited, Fire-Resistant, Synthetic Hydrocarbon Base
MIL-C-81302	-Cleaning Compound, Solvent, Trichlorotrifluoroethane
MIL-H-83282	-Hydraulic Fluid, Fire-Resistant, Synthetic Hydrocarbon Base, Aircraft

STANDARDS

FEDERAL

FED-STD-313	-Material Safety Data Sheets, Preparation and Submission of
FED-STD-791	-Lubricants, Liquid Fuels, and Related Products; Methods of Testing

MIILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-290	-Packaging, Packing, and Marking of Petroleum and Related Products

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply, except for the specific issue adopted by the Department of Defense as listed in the current Department of Defense Index of Specifications and Standards.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 93	-Flash Point by Pensky-Martens Closed Tester, Test for.
D 97	-Pour Point of Petroleum Oils, Test for.
D 130	-Copper Corrosion from Petroleum Products by the Copper Strip Tarnish Test, Detection of
D 270	-Petroleum and Petroleum Products, Sampling.
D 287	-API Gravity of Crude Petroleum and Petroleum Products (Hydrometer Method), Test for.
D 445	-Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity), Test for.
D 664	-Neutralization Number by Potentiometric Titration, Test for.
D 892	-Foaming Characteristics of Lubricating Oils, Test for.
D 972	-Evaporation Loss of Lubricating Greases and Oils, Test for.
D 1500	-ASTM Color of Petroleum Products (ASTM Color Scale), Test for.
D 1744	-Water in Liquid Petroleum Products by Karl Fischer Reagent, Test for.
D 2266	-Wear Preventive Characteristics of Lubricating Grease (Four-Ball Method), Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1816 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference form libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Qualification. The hydraulic fluid furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list at the time set for opening of bids, see 4.3.1 and 6.3. Any change in the formulation of an approved product shall require requalification.

3.2 Materials. The fluid shall consist of petroleum products with additive materials to improve the viscosity-temperature characteristics, resistance to oxidation, and antiwear properties of the finished product.

3.3 Petroleum base stock requirements. The properties of the petroleum base stock used in compounding the finished fluid, before the addition of any other ingredients required herein, shall be as designated in Table I when tested as specified in 4.7.2.

3.4 Additive materials.

3.4.1 Viscosity-temperature coefficient improvers. Polymeric materials may be added to the base petroleum oil in quantities not to exceed 20 percent by weight of active ingredient in order to adjust the viscosity of the finished fluid to the values specified in 3.5.

3.4.2 Oxidation inhibitors. Oxidation inhibitors shall be added to the base oil in quantities not to exceed 2 percent by weight.

3.4.3 Antiwear agent. The hydraulic fluid shall contain 0.5 ± 0.1 percent by weight of an anti-wear agent such as tricresyl phosphate, conforming to TT-T-656 or equivalent. When Tricresyl phosphate is used it shall contain no more than 1 percent of the ortho-isomer.

3.5 Finished fluid. The properties of the finished fluid shall be specified in Table II and 3.5.1 through 3.5.9.

3.5.1 Color. The fluid shall contain red dye in concentration not greater than 1 part of dye per 10,000 parts of oil by weight. There shall be no readily discernible difference in the color of the finished fluid and the standard color when tested as set forth in 4.7.3.

3.5.2 Corrosiveness and oxidation stability.

3.5.2.1 Corrosiveness. When tested as specified in 4.7.2 at a temperature of 135±1°C (275±2°F), the change in weight of steel, aluminum alloy, magnesium alloy, and cadmium-plated steel subjected to the action of the hydraulic fluid shall be not greater than ±0.2 milligrams

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ligrams per square centimeter of surface. The change in weight of copper under the same conditions shall be no greater than ±0.6 milligram per square centimeter of surface. There shall be no pitting, etching, nor visible corrosion on the surface of the metals when viewed under magnification of 20 diameters. Any corrosion produced on the surface of the copper shall be not greater than No. 3 of the ASTM copper corrosion standards (Method D-130). A slight discoloration of the cadmium shall be permitted.

3.5.2.2 Resistance to oxidation. When tested as specified in 4.7.2, at a temperature of 135±1°C (275±2°F) the fluid shall not have changed more than -5 or +20 percent from the original viscosity in centistokes at 40°C (104.0°F) after the oxidation-corrosion test. The acid or base number shall not have increased by more than 0.20 over the acid or base number of the original sample. There shall be no evidence of separation of insoluble materials nor gumming of the fluid.

3.5.3 Low temperature stability. When tested as specified in 4.7.2 for 72 hours at a temperature of -54±1°C (-65±2°F), the fluid shall show no evidence of gelling, crystallization, solidification, or separation of ingredients. Any turbidity shall be not greater than that shown by the turbidity standard.

3.5.4 Shear stability. When tested as specified in 4.7.4 the percent viscosity decrease of the hydraulic fluid, measured in centistokes at 40°C (104.0°F) and at -40°C (-40°F), shall be no greater than the percentage viscosity decrease of the shear stability reference fluid nor shall the acid or base number have increased by more than 0.20 over the original acid or base number.

3.5.5 Solid particle contamination.

3.5.5.1 Partical size. When samples taken for particle count are tested in accordance with 4.7.2 in a clean dust free atmosphere, the number of solid contamination particles per 100 ml of the fluid shall not exceed the number specified in Table III.

3.5.5.2 Gravimetric Method. The sample for solid particle contamination when tested in accordance with 4.7.5 shall not exceed 0.3mg/100ml.

3.5.5.3 Filtration time. The filtering time for each determination in 4.7.5 shall be 15 minutes maximum at 25 ± 5°C (77 ± 9°F).

3.5.6 Foaming characteristics. The foaming characteristics of the hydraulic fluid shall not exceed the limits indicated in Table IV when tested as specified in 4.7.2.

3.5.7 Storage stability. The fully blended product shall show no separation of ingredients nor evidence of crystallization, shall be clear and transparent when examined visually, and shall conform to the requirements of Section 3 after 12 months storage as specified in 4.7.2.

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3.5.8 Workmanship. The workmanship shall be in accordance with high-grade commercial practice covering this type of material. The finished fluid shall be homogeneous and free from suspended matter, grit or other adulteration.

3.5.9 Toxicity. The hydraulic fluid shall have no adverse effect on the health of personnel when used for its intended purpose. The fluid, shall contain no components which produce noxious vapors in such concentrations as to be an annoyance to personnel during formulation or use under conditions of adequate ventilation while exercising caution to avoid prolonged contact with the skin and while observing Occupational Safety and Health Administration (OSHA) guidelines. Questions pertaining to the toxic effects shall be referred by the procuring activity to the appropriate departmental-medical service who will act as an advisor to the procuring activity (see 4.4).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Governmnet reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and testing of the hydraulic fluid shall be classified as follows:

(a) Qualification inspection

(b) Periodic qualification re-evaluation

(c) Quality conformance inspection

4.3 Sampling. All sampling shall be conducted in a clean room environment.

4.3.1 Qualification samples. Qualification samples shall consist of three 1-gallon containers of hydraulic fluid, 1 gallon of the petroleum base stock before the addition of additive agents, 1 gram of the red dye, 8 ounces of the additive used for improving the viscosity-temperature coefficient, 8 ounces of the antiwear agent, and 1 ounce of the compound used for improving the oxidation stability. In the event that additives are supplied as concentrated solutions, an

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equivalent quantity of the solution shall be furnished. The samples shall be accompanied by a test report from the manufacturer or a commercial laboratory containing complete information as to the source and type of base stock and additive materials used, the formulation and composition of the finished fluid, and laboratory data showing quantitative results of all the tests required by this specification except storage stability. Separate qualification inspection shall be required for each base stock used. The samples and reports shall be forwarded to the Vehicle Technology Department, Naval Air Development Center (Code 60612), Warminster, Pennsylvania 18974. The samples shall be plainly identified by securely attached durable tags or labels marked with the following information:

Sample for qualification inspection
HYDRAULIC FLUID, PETROLUEM BASE; AIRCRAFT, MISSILE, AND ORDNANCE
MIL-H-5606D
Name of ingredient (for ingredient material)
Name of manufacturer
Product code number
Date of manufacture

Submitted by (name) (date) for qualification inspection in accordance with the requirements of MIL-H-5606D under authorization of (reference authorizing letter) (see 6.3).

4.3.1.1 Formulation sheets. An example of a satisfactory form for the formulation sheet, indicating the percentage and nature of each ingredient, is as follows:

Petroleum oil base stock (composition)	percent
Viscosity index improver (mfr's name and no.)	percent
Anti-wear additive (mfr's name and no.)	percent
Oxidation inhibitor (mfr's name and no.)	percent

4.3.2 Periodic qualification reevaluation samples. A periodic qualification reevaluation sample of two 1-gallon hermetically sealed cans of hydraulic fluid shall be selected at random from the first lot of fluid processed under a contract or order after the

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product has passed qualification inspection. Additional samples may be taken at intervals as considered necessary to verify the consistency of production quality. Periodic qualification reevaluation samples shall be forwarded to the laboratory responsible for qualification (see 6.3). The samples shall be plainly identified by securely attached durable tags or labels marked with the following information:

Sample for periodic qualification reevaluation
HYDRAULIC FLUID, PETROLEUM BASE; AIRCRAFT, MISSILE, AND ORDNANCE
MIL-H-5606D
Name of manufacturer
Product code number
Date of manufacture
Contract or order number
Batch number

4.3.3 Quality conformance inspection samples. The quality conformance inspection samples shall consist of a sample for tests (4.3.3.3), samples for examination of filled containers (4.3.3.4) and samples for determination of solid particle count (4.3.3.5). Samples shall be labeled completely with information identifying the purposes of the sample, name of product, specification number, lot and batch number, date of sampling, and contract number.

4.3.3.1 Bulk lot. A bulk lot (batch) is an indefinite quantity of a homogeneous mixture of material offered for acceptance in a single isolated container; or manufactured in a single plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material.

4.3.3.2 Packaged lot. A packaged lot is an indefinite number of 55-gallon drums or smaller unit containers of identical size and type, offered for acceptance, and filled with a homogeneous mixture of material from one isolated container; or filled with a homogeneous mixture of material manufactured in a single plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material.

4.3.3.3 Samples for tests. Take samples for tests in accordance with ASTM D270. The samples shall be subjected to all the applicable Quality conformance tests. If the samples for tests fail any of the Quality conformance tests, the inspection lot shall be rejected. In addition, a sample of base oil shall be selected in accordance

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with ASTM D270 for each lot of the finished fluid and shall be subjected to all the applicable Quality conformance tests for base oil.

4.3.3.4 Samples for examination of filled containers. Random samples of filled unit containers and a sample of shipping containers fully prepared for delivery shall be selected from each lot of fluid in accordance with MIL-STD-105 at inspection level II and acceptable quality level (AQL) = 2.5 percent defective.

4.3.3.5 Samples for determination of solid particle count. Samples of filled and sealed containers shall be taken at such periodic intervals as to be representative of each day's operation. The number of samples taken each day shall be in accordance with MIL-STD-105 at Inspection Level S-3. The sample size and number of determinations shall be as specified in Table V.

Should the particle count on any individual determination be considered excessive, two additional determinations on another sample from the same container may be used. The container shall be thoroughly shaken for approximately one minute immediately prior to withdrawing each 100-ml portion for all determinations. The arithmetic average of the two closer particle counts shall be considered the particle count for the sample.

4.4 Qualification inspection. Qualification inspection shall consist of a review for approval of the submitted manufacturer's report and subjecting the qualification samples (4.3.1) to examination and testing for all the requirements of this specification. Material Safety Data Sheets shall be prepared and submitted to the Qualifying Laboratory (4.3.1) in accordance with FED-STD-313.

4.5 Periodic qualification reevaluation. The periodic qualification reevaluation performed by the qualification laboratory shall consist of examining and testing the periodic qualification reevaluation sample for all the requirements of this specification except storage stability. Delivery of the fluid will not be delayed pending completion of this inspection. If the results of the periodic qualification reevaluation are in accordance with the requirements of this specification, the consistency of production quality shall be thereby verified. Failure of the fluid to pass a periodic qualification reevaluation shall require the acceptance and further shipment of the fluid to be withheld until the contractor has corrected the conditions which led to the failure (see 6.3.1). Further failure will constitute cause for rejection and removal from the Qualified Products List.

4.6 Quality conformance inspection. Quality conformance inspection shall consist of examination of the sample of filled containers,

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determination of the solid particle contamination count, and testing the sample for test for all the requirements specified in Section 3 except corrosiveness and oxidation stability, shear stability, swelling of synthetic rubber, and storage stability.

4.7 Inspection methods. Inspection shall be in accordance with Method 9601 of FED-STD-791.

4.7.1 Conformance of the fluid to the requirements for base stock (3.2) and additive materials (3.4) and finished fluids (3.5), shall be determined by appropriate examination and testing in accordance with 4.7.2.

4.7.2 Tests shall be performed in accordance with the applicable methods specified in Table VI, and 4.7.3 through 4.7.6. Physical and chemical values specified in Section 3 apply to the average of the determinations made on the samples for those values which fall within any stated repeatability or reproducibility limits of the applicable test method.

4.7.3 Color of finished fluid. The color of the hydraulic fluid shall be compared with a standard sample prepared by adding one part of dye, National Aniline and Chemical Company "Oil, Red O" to 10,000 parts of an oil not darker than ASTM D1500 Standard Number 1.

4.7.4 Shear stability. The shear stability shall be tested by subjecting the fluid to sonic oscillation.1/ The control fluid and the test fluid shall be run consecutively in the same apparatus and under the same test conditions. The test shall be of 30 minutes duration and the equipment so regulated that the viscosity decrease of the reference fluid (see 6.4) at 40°C (104.0°F) is approximately 15 percent. 30 ml of fluid shall be used in this test.

1/ A 250W, 10KC magnetostrictive oscillator Model DF-101 supplied by the Raytheon Corporation , or its equivalent, may be used. Directions in the manual for respective instruments shall be followed.

4.7.5 Solid particle contamination (gravimetric method). The gravimetric procedure for the solid particle contamination determination shall employ procedures, apparatus and materials similar to those used in FED-STD-791, Method 3009 or equivalent. As in Method 3009, the apparatus and material shall be carefully cleaned and solvents filtered so that the source of any solid particle contimination must be the test hydraulic fluid itself. This test shall be conducted in a clean, dust-free area with a relative humidity maintained below 50 percent. Two

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percent. Two white 47-mm cellulose membrane filters of 0.45 micron porosity shall be rinsed with filtered n-heptane distillate (95 percent minimum) 1/, placed using forceps in clean, loosely-covered containers, and dried to constant weight in a clean oven at approximately 80°C (176°F). The filters shall be allowed to reach equilibrium with the ambient test area temperature. The filters and their containers shall be placed on an air ionizer to remove any static charge and the filters then weighed to the nearest 0.01 mg to determine their tare weights. The control filter shall be placed below the sample filter in the filter holder, 100 ± 1 ml of the hydraulic fluid, which has been agitated for approximately 1 minute, shall be filtered using a vacuum of 14 to 17 inches of mercury and washed with 200 ml of filtered n-heptane distillate (95 percent minimum) 1/, from a wash bottle. The last 50 ml of solvent shall be used to wash the test fluid from the periphery of the filter after the upper half of the filter holder has been removed. The solvent flow shall be directed gently towards the center of the filter. The sample filter and the control filter shall again be placed in clean, loosely-covered containers and dried to constant weight in an oven at approximately 80°C (176°F). The filters shall be allowed to come to equilibrium with the ambient test area temperature; the containers with filters placed on an air ionizer, and the filters then weighed to the nearest 0.01 mg. The weight change of the sample filter shall be corrected by the weight change of the control filter. The weight of contaminate shall be expressed in mg per 100 ml of fluid sample. The two filters shall be examined under a microscope of approximately 35X magnification. The presence of large or unusual particles shall be cause for discarding and another test shall be made.

1/ MIL-C-81302 Type I (1, 1, 2 trichlorotrifluoroethane) may be substituted if flammable liquids are unacceptable. Petroleum ether is permissible where n-heptane is not readily available.

4.7.6 Copper strip corrosion. Prepare 3 copper strips in accordance with ASTM D130. The copper strips shall be placed individually in test tubes or other suitable containers so that they will be immersed by 90 ml of the hydraulic fluid. Each fluid container, equipped with an air condenser, shall be immersed in a constant temperature bath capable of maintaining the fluid temperature at 135 ±1°C (275 ±2°F). After 72 hours at this test temperature, the strips shall be removed from the fluid, rinsed in sulfur-free acetone, and compared with the ASTM Copper Strip Corrosion Standards.

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4.7.7 Examination of filled containers. Each sample of filled container and shipping container shall be examined for defects of construction of the container and closure, evidence of leakage, and net content. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for acceptance inspection provided that the contractor has removed or repaired all non-conforming containers.

5. PACKAGING

5.1 Packaging and packing. The packaging and packing of the hydraulic fluid shall be in accordance with MIL-STD-290. The fluid shall be furnished in 8 ounce, 1-quart, 1-gallon and 10-gallon metal cans and 55-gallon drums as specified (6.2). The quart and gallon cans shall conform to Type I of PPP-C-96. The 10 gallon metal pails shall conform to Type I, Class II of PPP-P-704 except that all dimensions shall be in accordance with Figure 1. Plugs and flanges shall be in accordance with PPP-P-420:

3/4-inch flange	Type I, Class B
3/4-inch plug	Type I. Class D
2 -inch flange	Type I, Class A
2 -inch plug	Type I, Class C

The interior of the pail shall be smoothly and continuously covered to a minimum thickness of 0.001 inch with two well-cured coats of an epoxy-phenolic container coating resin. The coating resin shall be compatible with MIL-H-5606 and shall neither release particles nor flake off during normal use of pail, including minor denting and creasing of drum surfaces from external impact. The gasket material used shall be compatible with MIL-H-5606 and shall not have any adverse effect on the hydraulic fluid. Just prior to filling, all containers shall be thoroughly cleaned, rinsed with clean filtered fluid, and examined to insure absolute absence of loose solder dirt, fibers, lint, metal particles, seaming compound, corrosion products, water, or other foreign contaminants.

5.2 Marking. The marking of all containers shall be in accordance with MIL-STD-290, and any special marking required in the contract or order (6.2). The unit containers shall also be marked with the following information:

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DESTROY THIS CONTAINER WHEN EMPTY. THIS FLUID IS COMPATIBLE WITH MIL-H-6083, MIL-H-83282 AND MIL-H-46170. DO NOT USE WITH ANY OTHER TYPE OR GRADE OF HYDRAULIC FLUID.

6. NOTES

6.1 Intended use. The hydraulic fluid covered by this specification is intended for use in automatic pilots, shock absorbers, brakes, flap-control mechanisms, missile hydraulic servo-controlled systems, and other hydraulic systems using synthetic sealing material.

6.1.1 Storage conditions. Prior to use in the intended equipment, the product may be stored under conditions of covered or uncovered storage in geographic areas ranging in temperatures from -57° to +49°C (-70° to +120°F).

6.1.2 This fluid is not interchangeable with any other type or grade of hydraulic fluid except MIL-H-6083, MIL-H-83282 and MIL-H-46170.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Type and size of containers (see 5.1)

c. Quantity

d. Selection of applicable levels of packaging and packing with requirements in detail, (see 5.1)

e. Special marking (see 5.2)

f. Toxicological data requirements (3.2 and 4.4)

6.2.1 Basis of purchase. The fluid covered by this specification should be purchased by volume, the unit being a U.S. gallon of 231 cubic inches at 15.6°C (60°F).

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6.2.2 List of qualified products. Products considered acceptable under this specification are listed in QPL-5606 and subsequent revisions thereto.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Naval Air Systems Command, Department of the Navy; however, information pertaining to qualification or products may be obtained from the Vehicle Technology Department, Naval Air Development Center, (Code 60612) Warminster, Pennsylvania 18974.

6.3.1 It is understood that the material furnished under this specification subsequent to final approval shall be of the same composition and shall be equal to products upon which approval was originally granted. In the event that the fluid furnished under contract is found to deviate from the composition of the approval product, or that the product fails to perform satisfactorily, approval of such products will be subject to immediate withdrawal from the Qualified Products List.

6.4 Samples of the standard synthetic rubber L for the test specified in 4.7.2 and 1 pint of the reference fluid for the test specified in 4.7.4 may be obtained from the Vehicle Technology Department, Naval Air Development Center, (Code 60612) Warminster, Pennsylvania 18974.

6.5 The provisions of 1.1 of this specification are the subject of international standardization agreement (ASCC 15/1, STANAG 1135 and STANAG 3748). When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Office (DepSO) so that appropriate action may be taken respecting the international agreement concerned.

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6.6 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - MR
Navy - AS
Air Force - 11

Review Activities:
Army - AV, MI, EA, ME, AR
Navy - SH
Air Force - 68

User Activities:
Navy - MC, OS

Preparing Activity:
Navy - AS

(Project No. 9150-0492)

International Interest:
(see Section 6)

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TABLE I. Properties of petroleum base stock. (see 3.3)

Property	Limits
Pour point (max) 1/	-59.4°C (75.0°F)
Flash point (min)	82°C. (179.6°F)
Acid or base No. (max)	0.10
Color, ASTM Std (max)	No. 1
Specific gravity at 15.6°C/15.6°C (60.0°F/60.5°F)	Report 2/

1/ Pour point depressant materials shall not be used.

2/ Samples of base stock submitted for acceptance tests shall not vary by more than ± 0.008 from the specific gravity of the original sample submitted for qualification tests.

TABLE II. Properties of finished fluid. (see 3.5)

Property	Limits	Test paragraph
Viscosity in centistokes at 100°C. (212°F) (min).	5.0	4.7.2
Viscosity in centistokes at 40°C. (104°F). (min).	13.0	4.7.2
Viscosity in centistokes at -40°C (-40°F.) (max).	500	4.7.2
Viscosity in centistokes at -54°C (-65°F.) (max).	3000	4.7.2
Pour point (max) 1/	-59.4°C. (-75.0°F.)	4.7.2
Flash point (min)	82°C (179.6°F.)	4.7.2
Acid or base No. (max)	0.20	4.7.2
Rubber Swell, Standard Synthetic rubber L, percent	19.0-30.0	4.7.2
Evaporation loss, percent, (max)	20	4.7.2
Copper Strip Corrosion, ASTM Standard, (max)	No. 2e	4.7.6
Water, parts per million total, (max)	100	4.7.2
Steel-on-Steel wear (average wear scar), millimeters in diameter, (max)	1.0	4.7.2

1/ Pour point depressant materials shall not be used.

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TABLE III. Solid contaminant particles.
(See 3.5.5)

Particle size range (largest dimension), microns	Allowable number (max) each determination	
	Microscopic count	Automatic count
5-15	2500	10,000
16-25	1000	1,000
26-50	250	150
51-100	25	20
over 100	10	5

TABLE IV. Foaming characteristics of hydraulic fluid. (see 3.5.6)

Test temperature	Foaming tendency	Foam stability
	Foam volume, ml, at end of 5-minute blowing period	Foam volume, ml, at end of 10-minute settling period
24°C (75°F)	65 ml (max)	Complete collapse 1/

1/ A ring of small bubbles around the edge of the graduate shall be considered complete collapse.

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TABLE V - Sample for particle contamination. (see 4.3.3.5)

Container	Sample size (ml)[1/]	Number of determinations per sample
8 ounce	100	1
1 quart	100	1
1 gallon	200	2
5 gallon	300	3
10 gallon	300	3
55 gallon	600	6

1/ Each determination shall be made on 100-ml portions of the sample.

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TABLE VI. Inspection methods. (see 4.7.2)

Test	FED-STD-791 Method No.	ASTM Method No.
Specific Gravity		D287
Viscosity		D445
Pour Point		D97
Flash Point		D93
Acid or Base No		D664
Color		D1500
Corrosiveness and Oxidation Stability	5308	
Low Temperature Stability	3459	
Swelling of Synthetic Rubber	3603 1/	
Evaporation		D972 2/
Solid Particle Contamination	3009 3/, 4/	
Foaming Characteristics		D892
Water		D1744
Storage Stability	3465	
Steel-on-Steel Wear		D2266

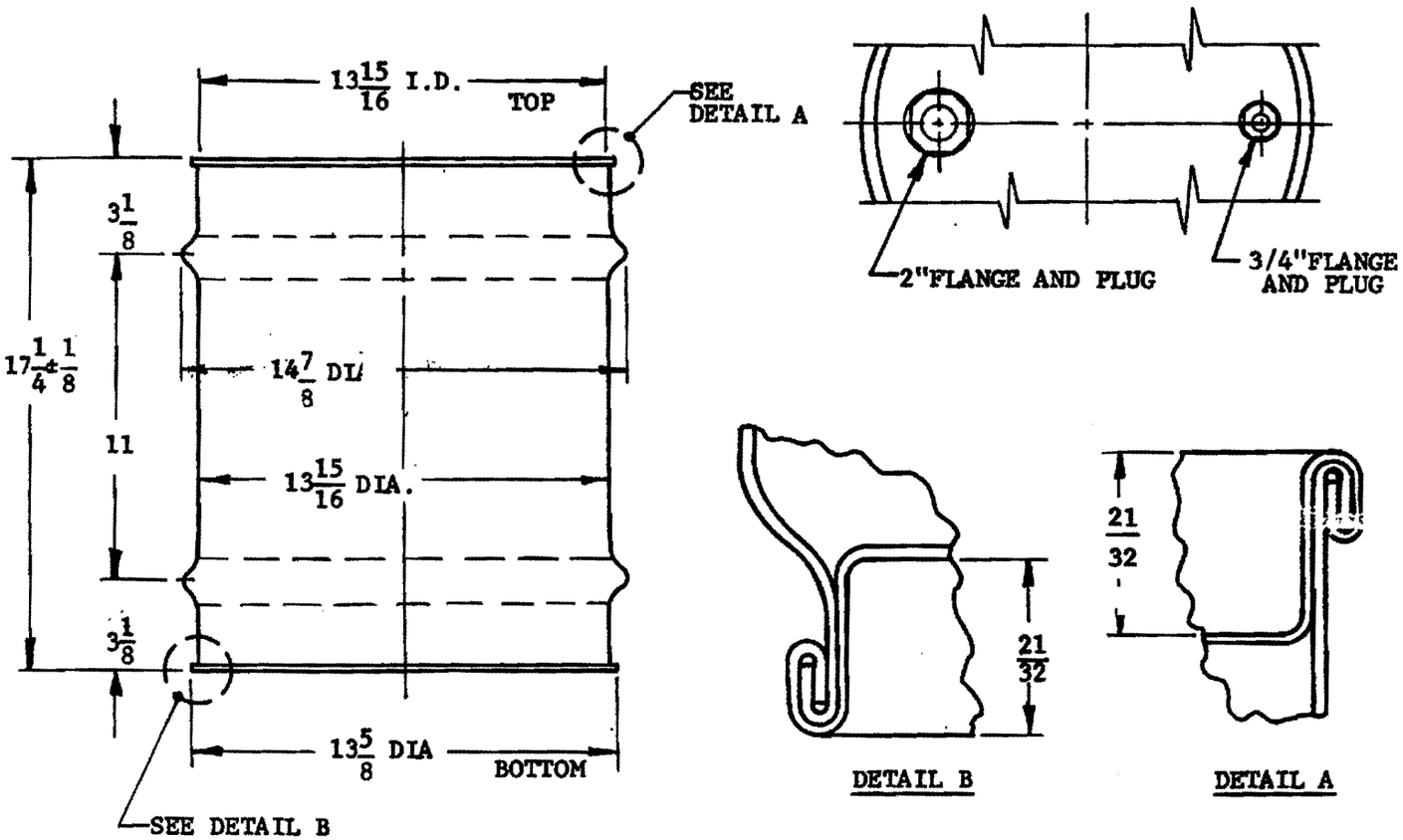
1/ Use standard rubber L (see 6.4) within 6 months after date of manufacture.

2/ Test shall be run for 6 hours at 71°C (159.8°F).

3/ Two 100-ml increments of n-heptane distillate (95 percent minimum) shall be used to wash the sample. Normal heptane shall be substituted for petroleum ether when testing for solid particle contamination.

4/ Particle contamination shall be measured by either the use of automatic particle counters or the optical procedure detailed in Method 3009. HiAc counter Model PC-203, PC-202, or equivalent counting to the limits specified in Table III may be used. Directions in the manual for respective instruments shall be followed. The Hiac counter method shall be the reference method used in resolution of dissimilar test results from different laboratories.

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DIMENSIONS IN INCHES
MATERIAL - STEEL
TOLERANCES (EXCEPT WHERE INDICATED): ±1/16

FIGURE 1. 10 gallon pail.

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